LLM-Driven Unified Framework with Grid Diffusion for Efficient Long Video Editing

Deyin Liu[a], Li He[a], Zhan Li[b], Chengwu Liang[c]

[a]*Engineering Research Center of Autonomous Unmanned System Technology, Ministry of Education; Anhui Provincial Key Laboratory of Security Artificial Intelligence; and School of Artificial Intelligence, Anhui University, No 111, Jiulong Road, Hefei, 230601, China*
[b]*The Institute of Smart City and Intelligent Transportation, Southwest Jiaotong University, No 999 Xi'an Road, Chengdu, 611756, China*
[c]*Henan University of Urban Construction, Longxiang Road, Pingdingshan, 467036, China*

Abstract

Text-guided video editing aims to transform a source video according to a target text prompt while preserving desired visual content and temporal coherence. Despite recent progress, existing approaches often target specific editing scenarios (e.g., zero-shot or one-shot, local or global editing), struggle to process long videos due to memory and computational constraints. In this paper, we propose UELVE, an Large Language Model-driven Unified framework for Efficient Long-Video Editing based on pre-trained text-to-image diffusion models. UELVE first exploits the reasoning capability of an LLM to automatically analyze the source and target prompts, enabling adaptive decisions on editing strategies. To support long-form video editing, we organize video frames into grids and perform sequential grid-based diffusion, where grid self-attention explicitly ensures short-term consistency among neighboring frames. Furthermore, a simple yet effective grid sliding mechanism is introduced to establish inter-grid interactions, maintaining long-term temporal consistency. In addition, UELVE naturally supports multi-GPU parallel inference with only minimal inter-GPU communication, substantially improving editing efficiency. Extensive experiments

on a wide range of video editing scenarios demonstrate that UELVE consistently achieves superior editing quality, strong temporal consistency, and significantly improved efficiency compared with state-of-the-art methods.
Keywords: Large Language Models, Grid diffusion, Efficient long video editing

1. Introduction

Text-guided video editing [1] aims to transform a source video into a new one in a generative manner, such that the resulting video semantically conforms to a target text prompt while simultaneously preserving some information from the source video as needed. In recent years, building upon publicly available pre-trained Text-to-Image (T2I) diffusion models [2, 3], text-guided image editing has advanced substantially [4]. In the video domain, a series of large-scale Text-to-Video (T2V) generative models have emerged, including close-sourced [5, 6] and open-sourced [7, 8]. However, existing video diffusion models are generally vast in volume [9], especially, for long-form video generation and editing [1], current T2V models suffer from limited efficiency and flexibility due to memory and computational bottlenecks, although some methods claim that their models are extensible to handle longer video sequences [10, 11, 12]. Thus, for text-guided video editing tasks, many existing works instead adapt pre-trained T2I diffusion models to video data by additionally considering temporal-awareness information. These methods generally can be divided into two categories: zero-shot and one-shot video editing approaches. The former works [13, 14, 15], in a training-free manner, are particularly suitable for simple editing tasks that involve loose inter-frame consistency or less temporal continuity requirements such as style transfer and color tone adjustment, while the latter [16, 17] emphasize on learning close across-frame correlation or temporal-awareness infor-

mation via per-video fine-tuning, enabling more complex edits such as motion preservation or modification. Besides, no matter zero-shot or one-shot setting, different methods may be adept at doing editing of different spatial scales (local or global).

Currently, several challenges exist in this area. **Firstly**, there is still no unified framework that can flexibly handle diverse video editing tasks (zero-shot or one-shot, local or global) according to the specific circumstances. **Secondly**, many existing T2I-based video editing methods are limited to handling videos with at most several dozens of frames [14, 18, 19]. They often perform diffusion operations on an entire video bulk [16, 17], which severely restricts the length of videos that can be processed due to the computational limitation of the GPU. However, preliminary attempts in extending the length of the edited video inevitably compromises the editing performances [20, 21] because of the inability of simultaneously keeping short-term and long-term consistencies. **Thirdly**, the editing speeds of existing methods remain relatively slow. Although recent studies have explored various strategies (including token merging [18], caching and reuse [22], and parallel computation [23, 24]) to accelerate diffusion model inference, the efficiency of long-video editing still falls short of the requirements for real-world applications.

To address the aforementioned challenges, we propose a Large Language Model (LLM)-driven **U**nified framework, which performs flexible grid manipulations based on a pre-trained T2I diffusion model to enable **E**fficient **L**ong-**V**ideo **E**diting across a wide range of editing tasks, termed as **UELVE**. Representative examples edited by UELVE are illustrated in Figure 1, including object/background replacement, local attribute modification, global stylization, as well as motion preservation. We first exploit the reasoning capability of a LLM



Figure 1: **UELVE** is an efficient long-video editing method that leverages LLM to make decision on model adaptions according to specific editing types, implement grid diffusion on a pre-trained Text-to-Image model to maintain short-term and long-term consistencies, and enhance the efficiency by multi-GPU data parallel computation. It is applicable to diverse editing tasks, including object/background replacement, local editing of attributes (style, color, shape) , global style transformation and even motion editing.

to compare and analyze the source and target text prompts so as to identify the specific video editing goals and types. The LLM could serve as an agent to automatically make wise decisions between zero-shot and one-shot settings (involving whether per-video fine-tuning [25] is required or not), as well as between local and global editing (involving whether spatial masks are needed or not). This design establishes the foundation for a unified, more precisely targeted, and cost-effective video editing framework.

Then, inspired by the grid trick (also known as the character sheet), a commonly used hack in the Stable Diffusion art community [2], we arrange the input video frames into a sequence of grids (e.g., each grid is with dimension 2×2, accommodating four frames) and perform T2I diffusion sequentially by inputting one grid every time. Thus, UELVE is able to leverage *grid self-attention* to explicitly facilitate neighbor-frame interactions within each grid, thereby ensuring short-term consistency. To further maintain long-term consistency, we introduce a simple yet effective *grid sliding manipulation*, which is applied to both the video inversion and denoising diffusion processes to enable inter-grid interactions. Based on such operations, UELVE decomposes a long video into manageable batches, enabling to flexibly edit videos of arbitrary length in theory using only a single GPU in a memory-swapping manner.

Furthermore, UELVE also supports multi-GPU parallelization to accelerate the editing process, while requiring only a small amount of data communication (typically a few frames near the data division boundaries) between GPUs. This design not only reduces the overhead caused by large-scale inter-GPU communication (e.g., if data shuffling proposed in [14] is used on multiple GPUs), but also maintains across-GPU consistency.

Extensive experiments demonstrate both quantitatively and qualitatively that UELVE achieves effective and efficient long-form video editing across diverse editing tasks. To summarize, our main contributions are as follows:

- We propose a novel unified diffusion framework for video editing, which leverages the reasoning capability of LLMs to make adaptive decisions on model adaptions according to the specific editing goals and types.

- By organizing video frames into grids for batched processing, we intro-

duce grid self-attention based on a pre-trained T2I diffusion model to enhance short-term consistency across intra-grid neighboring frames. We also design a grid sliding manipulation to establish inter-grid connections during the diffusion process, maintaining long-term consistency and thus enabling editing of long videos.

- To enhance the efficiency, UELVE can be readily adapted to multi-GPU parallel computation to accelerate the editing process while maintaining cross-GPU consistency, with only minor inter-GPU data communication.

- UELVE is designed as a plug-in framework that can be seamlessly integrated with various existing image editing approaches, such as attention control [26] or ControlNet-based methods [27], to achieve diverse video editing effects.

2. Related Work

2.1. Text-guided Video Generation and Editing

Recently, open-sourced video generation models (such as [8, 28]) pretrained on large-scale text-video pairs have emerged, which boosts significantly the applied research of Text-to-Video (T2V) models. However, due to the large size of these models, the memory requirements are extremely high, and for long-form video applications, they are inflexible and more memory-cost, which is impractical in memory-resource-limited environments. Therefore, many existing approaches to video generation and editing resort to the popular large-scale pretrained Text-to-Image (T2I) diffusion models [2], by adapting them to video data and considering temporal consistency.

Currently, T2I models based video generation and editing methods generally fall into two categories: zero-shot and one-shot approaches. Zero-shot methods [11, 13, 14] attempt to generate or edit videos in a training-free manner, but only cope well with simple videos that demand less in the temporal dimension. One-shot approaches [16, 29] use a video to fine-tune the pretrained model to achieve better temporal consistency and enable more complex video creation, but per-video fine-tuning is still costly. Besides, in terms of video editing, most of existing methods do not specially claim which types of editing (such as global or local) they are better at. Actually, these methods each have their own advantages and disadvantages. For examples, ControlNet [27] based methods generally do better at global style control, while [30] is specially designed for local video editing. In a nutshell, there is still no unified framework that can unify diverse video editing tasks.

LLM-Driven Video Editing: Large Language Models (LLMs) have recently been introduced to assist visual generation and editing tasks, however, existing methods primarily leverage LLMs to generate instructions and/or target prompts for constructing paired or triplet data to train instruction-based image/video editing models [31, 32, 33]. In contrast, our UELVE leverages LLM to analyze the changes between source and target prompts to make appropriate choices for the model adaptions according to the need of different editing tasks, which helps enhance the cost-effectiveness and practicality of the united video editing framework.

Long-form Video Editing: Existing video generation and editing methods like [34] perform fully cross-frame interactions among all video frames, at the cost of substantial computational overhead. Some others adopt sparse causal attention [16], frame attention [17], or spatiotemporal self-attention [15] to model

interactions between every frame and a single selected reference frame (the first, the immediately previous, or the middle frame), which however leads to insufficient modeling of inter-frame correlations. These approaches do not suit for long-form video editing. With the auto-regressive strategy, some methods [10, 11, 12] try to extend the method for short videos to long-form video editing, but it is hard to guarantee the quality of the edited videos.

2.2. Grid Diffusion

Diffusion models [35, 36] have demonstrated superior capabilities in visual generation. Recently, Taegyeong Lee et.al [37] first proposed a simple but effective novel grid diffusion for text-to-video generation without temporal dimension in architecture, which can generate a high-quality video using a fixed amount of GPU memory regardless of the number of frames by representing the video as a grid image. Then Maomao Li et.al presented Qffusion [38], a dual-frame-guided framework for portrait video editing. They consider a design principle of "animation for editing", and train Qffusion as a general animation framework from two still reference images while they can use it for portrait video editing easily by applying modified start and end frames as references during inference. Unlike these two methods leveraging interpolation or propagation to fill (generate) the missing frames, we will exploit flexible inter-grid interactions to enhance the long-term consistency of generated videos.

2.3. Diffusion Model Acceleration

Existing researches on diffusion model acceleration have focused mainly on reducing the number of sampling steps [39], designing optimal architectures [40], or leveraging feature merging [18], caching and reuse [22]. Yet, these methods often require additional training or fail to deliver strong acceleration in

practice, exhibiting a clear trade-off between generation quality and speed. Some parallel computation based methods [23, 24]) explored both data and model parallelism for diffusion models, but they still need large inter-GPU communication. In contrast, the unique grid manipulations enable our UELVE to leverage minor cross-GPU data communication to achieve desirable long-form video quality while accelerating the inference significantly.

3. Methodology

3.1. Overview

Let $\mathcal{V}_K = \{\mathcal{I}^1, \ldots, \mathcal{I}^K\}$ be an input video containing K frames, which is paired with a source prompt $\mathcal{P}$, and $\mathcal{I}^k \in \mathbb{R}^{W \times H \times C}$ denote the k-th frame in $\mathcal{V}_K$ where W, H and C specify the width, height and channel, respectively. Given a target prompt $\mathcal{P}^*$, the video editing task is to translate the source video $\mathcal{V}_K$ into a modified version $\mathcal{V}_K^*$, which is semantically aligned with $\mathcal{P}^*$ while preserving some basic information of $\mathcal{V}_K$ as needed. In this work, based on a LLM-driven unified framework with grid manipulations on a pre-trained T2I diffusion model, we aim for efficient long video editing covering diverse specific editing goals and types.

Our proposed UELVE framework is shown in Figure 2. We first arrange the frames of input video into grids, encode the grids via Variational AutoEncoder (VAE) [41] conforming to the paradigm of Latent Diffusion Models [2], and then run DDIM inversion [36] via T time steps of iterations, conditioned on the source prompt, to acquire the inverted noisy latent grids. Next, starting from the inverted noisy latents, we run grid diffusion denoising conditioned on the target prompt for T steps of iterations as well. The latent grids obtained at the last time step

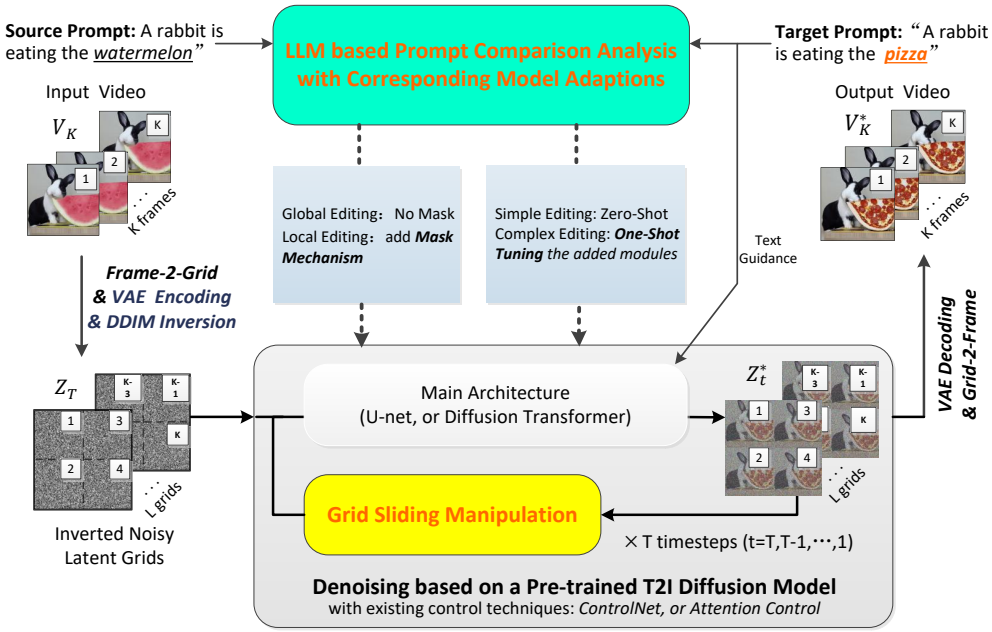


Figure 2: **Framework of UELVE**. We first arrange the frames of input video $\mathcal{V}_K$ into grids, encode them into the latent space and then invert the original latents to Z_T. Then we denoise the grid latents Z_T to Z_0^* via T time steps of iterations on a pretrained Text-to-Image diffusion model, decode Z_0^* and rearrange the resulted grid frames into an edited video $\mathcal{V}_K^*$ as output. For the denoising process, by comparing the source and target prompts, we leverage a LLM to reason out specific editing goals and types and thus make decisions on corresponding model adaptions, unifying diverse editing tasks into a same framework. Beyond the grid self attention for within-grid interactions, we perform a grid sliding manipulation to enhance cross-grid interactions to maintain long-term consistency. Benefiting from the flexible grid manipulations, a multi-GPU data parallel computing strategy is also considered to remarkably improve the computation efficiency. For simplicity, the control schemes from existing image editing approaches are omitted.

go through VAE decoding and then are rearranged into pixel frames chronologically, forming the edited video. It's worth noting that, for the diffusion process based on the pretrained T2I model, beyond the traditional text prompt guidance, we leverage a LLM to compare the source and target text prompts, and analyze

the specific editing goals and types, thus reason and make decisions on zero-shot or one-shot setting, local or global editing depending on whether fine-tuning is needed or not, whether mask is needed or not, respectively. Our framework could do corresponding model adaptions according to specific circumstances, such as integrating fine-tuning techniques for one-shot setting, or introducing masking mechanisms for local editing (Section 3.3). Besides, the grid self-attention make it possible to facilitate the neighbor-frame interactions within each grid to maintain the short-term consistency, on the other hand, to ensure the long-term consistency, we perform a simple yet effective grid sliding manipulation to achieve inter-grid interactions (Section 3.4). Moreover, our grid-based diffusion models are easy to be deployed on multiple GPUs to enable flexible data parallel computing, and by the grid sliding near the data division boundaries, only very little inter-GPU data communication is needed to maintain the cross-GPU consistency, which accelerates the diffusion process significantly (Section 3.5).

3.2. Preliminaries

Latent Diffusion Models (LDMs). LDMs [2] typically run the diffusion models (such as denoising diffusion probabilistic models (DDPM) [35]) in a lower-dimensional latent space produced by a pre-trained autoencoder, Variational AutoEncoder (VAE) [41], to significantly reduce the resources required for pixel-level calculation. In specific, the encoder $\mathcal{E}$ of VAE is used for compressing images (or video frames) x into latent representations $z = \mathcal{E}(x)$, and in the end of all the diffusion operations the decoder $\mathcal{D}$ reverts the latent (e.g., z^*) to pixel space, i.e., $\mathcal{D}(z^*) = x^*$. In the latent space, given the data $z_0 \sim q(z_0)$, the forward diffusion process is subject to a Markov Chain, which progressively corrupts z_0 to get $z_1, ..., z_T$. The Markov transition $q(z_t|z_{t-1})$ is defined as a Gaussian distri-

bution with a variance schedule $\beta_t \in (0,1),\ t = 1, ..., T$, that is,

$$q(z_t|z_{t-1}) = \mathcal{N}(z_t;\ \sqrt{1-\beta_t}z_{t-1}, \beta_t I) \tag{1}$$

According to Markov property, the conditional probabilities $q(z_t|z_0)$ can be acquired:

$$q(z_t|z_0) = \mathcal{N}\Big(z_t;\ \sqrt{\overline{\alpha}_t}z_0, (1-\overline{\alpha}_t)I\Big) \tag{2}$$

where $\alpha_t = 1 - \beta_t$, $\overline{\alpha}_t = \prod_{i=1}^{t} \alpha_i$. While in the backward denoising process, the model is to turn the noise into the original data distribution gradually. This process can be described as follows:

$$\begin{aligned} p_\theta(z_{t-1}|z_t) &= \mathcal{N}\Big(\mu_\theta(z_t, t), \Sigma_\theta(z_t, t)\Big) \\ &= \mathcal{N}\Big(z_{t-1}; \frac{1}{\sqrt{\alpha_t}}\Big(z_t - \frac{\beta_t}{\sqrt{1-\overline{\alpha}_t}}\epsilon_\theta(z_t, t)\Big), \frac{1-\overline{\alpha}_{t-1}}{1-\overline{\alpha}_t}\beta_t I\Big) \end{aligned} \tag{3}$$

where $\epsilon \in \mathcal{N}(0, I)$ and θ denote the parameters of the denoising neural network. In training stage, the objective is to maximize the likelihood of observed data $p_\theta(z_0) = \int p_\theta(z_{0:T})dz_{0:T}$ by maximizing its evidence lower bound (ELBO). The network is trained to realize the parameterized $p_\theta(z_{t-1}|z_t)$ to approximate the theoretical denoising process $q(z_{t-1}|z_t, z_0)$ which can be derived theoretically via Bayes' rules. Through simplification, the final objective is to train a network ϵ_θ to predict an artificial noise:

$$\min_\theta E_{z_0, \epsilon\sim\mathcal{N}(0,I), t\sim Uniform(1,T)}\|\epsilon - \epsilon_\theta(z_t, t, C)\|_2^2 \tag{4}$$

where z_t is a noised sample and the noise is added to the clean data z_0 according to timestamp t, and $C = \Phi(\mathcal{P})$ is the embedding of the text prompt condition which can be injected into $\epsilon_\theta(\cdot)$. At inference, given a random noisy vector z_T, the noise can be progressively removed by sequentially predicting $\epsilon_\theta(z_t, t, C)$ using the trained network for T steps. That is, by iteratively sampling z_{t-1}, the denoising process can generate a new sample z_0^* finally.

Denoising Diffusion Implicit Models (DDIM). DDIMs [36] are variants of DDPM [35], which can accelerate the denoising (sampling) process, belonging to a more efficient class of iterative implicit probabilistic models with the same training procedure and objective as DDPMs. The DDIM denoising is expressed as follows:

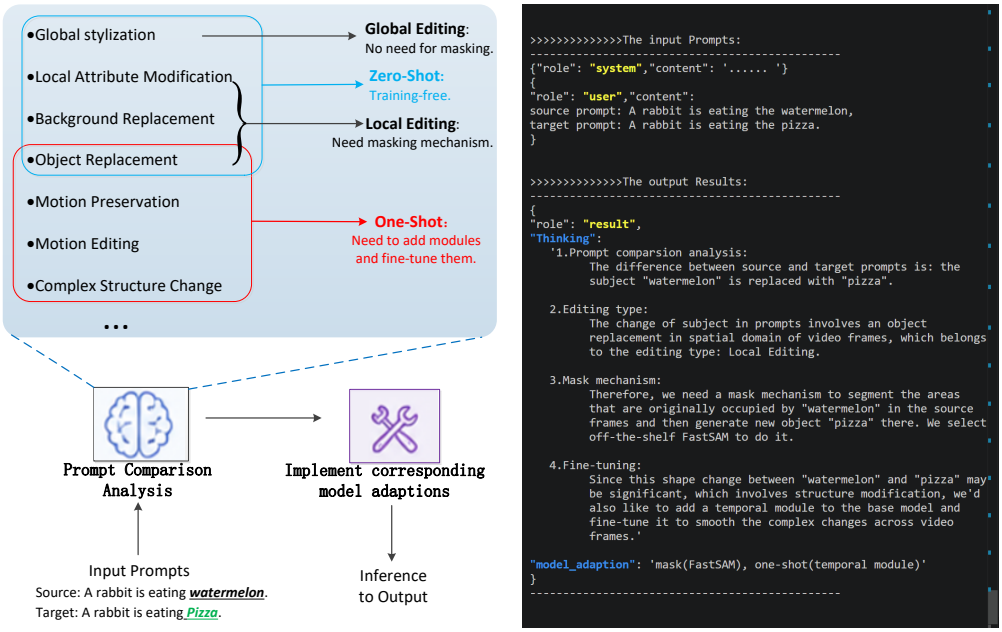
$$z_{t-1} = \sqrt{\frac{\alpha_{t-1}}{\alpha_t}} z_t + \sqrt{\alpha_{t-1}} \left(\sqrt{\frac{1}{\alpha_{t-1}} - 1} - \sqrt{\frac{1}{\alpha_t} - 1} \right) \epsilon_\theta(z_t, t, C) \tag{5}$$

In contrast to DDPMs, the denoising generative process of DDIMs is deterministic, which is a desirable property in the editing for real image/video. A simple inversion can be derived based on the ordinary differential equations (ODEs) limit analysis of the diffusion process, a.k.a. DDIM inversion, which can map a clean latent z_0 (encoded from real pixel data via VAE) to a noised latent $\hat{z}_T$ in limited time steps:

$$z_{t+1} = \sqrt{\frac{\alpha_{t+1}}{\alpha_t}} z_t + \sqrt{\alpha_{t+1}} \left(\sqrt{\frac{1}{\alpha_{t+1}} - 1} - \sqrt{\frac{1}{\alpha_t} - 1} \right) \epsilon_\theta(z_t, t, C) \tag{6}$$

3.3. LLM-driven Unified Diffusion Framework

Existing pretrained T2I models based video editing methods each have their respective strengths and weaknesses with each excelling in a limited set of categories of editing tasks. There exists still no unified framework that can combine the advantages of them to handle diverse video editing tasks. Therefore, we consider to leverage a LLM to help form a unified video editing framework based on a pretrained T2I diffusion model. Specifically, the LLM is first utilized to compare and analyze differences between the source and target text prompts. As shown in Figure 3 a), analyzing changes of words, LLM can automatically determine whether the editing belongs to global stylization or local editing (such as local attribute modification, background or object replacement), according to the



a) Schematic diagram of LLM based analysis and implementation

b) An example of the analysis process

Figure 3: **The illustration of LLM based prompt analysis and corresponding model adaptions**. The source and target text prompts are input into the LLM, which performs semantic comparison and analysis, thus infers the editing scope (local or global) and the degree of complexity (zero-shot or one-shot settings), and then recommends implementing corresponding model adaptions (such as adding masking mechanisms or modules with fine-tuning).

scope of the changes involved in every frame. Besides, in terms of the editing complexity, estimating whether or not the changes involve motion preservation or modification, or complex structure change, LLM determines the T2I model needs one-shot tuning or not. Then based on the analysis, our UELVE can implement various corresponding model adaptions on the pretrained T2I model. For example, if the editing type is local editing, UELVE incorporates an off-the-shelf masking mechanism (such as FastSAM [42]) to restrict the editing areas, and if motion preservation or modification is needed, UELVE adds a temporal

module and finetune it (as in [16, 19]), or if a big shape change occurs, UELVE just finetunes the inflated spatial-temporal attention (as stated in [15]) to maintain the consistency of new shape. In a nutshell, the LLM serves as an agent, which is guided to do smart comparison and analysis by being prompted with roughly pre-defined editing type classification guidelines as shown in Figure 3 a). Then various existing modules can be integrated for model adaptions as needed. An example of the analysis process for model adaption is show in Figure 3 b).

3.4. Grid Manipulations

Coupled with LLM-driven specific model adaptions and existing image editing approaches, UELVE has already been able to fulfill various editing tasks for short videos. In computation-resource limited environment, to cope with longer videos, we introduce gird diffusion to simultaneously maintain short-term and long-term consistencies. Specifically, we arrange the input video frames into a sequence of grids each with dimension $n \times l$ (e.g., 2×2, accommodating four frames) and perform T2I latent diffusion sequentially by inputting one grid every time. In such a way, the $N = nl$ frames in every latent grid Z_t will collectively go through the self attention module of the pretrained T2I model, which naturally enhances the consistency among them. Unlike existing methods, which perform either prohibitively expensive cross-all-frame interactions [10], or insufficient local neighbor-frame interactions (by sparse causal attention [16], frame attention [17], or spatiotemporal self-attention [15]), our UELVE leverages grid self-attention to facilitate neighbor-frame interactions within each grid to effectively ensure short-term consistency.

Furthermore, when confronting with long videos, although guided by the same text prompt, the diffusion trajectories of different latent grids may diverge

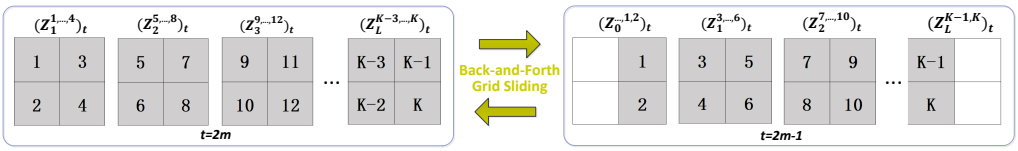


Figure 4: **The illustration of grid sliding manipulation** between odd-even time steps of the diffusion process.

over the time steps, leading to disparities across grids. To address this, we perform a grid sliding manipulation in the diffusion processes to achieve cross-grid interactions, thus maintaining long-term consistency. To be specific, for a video of K frames, in the even time steps of diffusion processes, we arrange the latent frames chronologically into L grids, forming the following grid layout:

$$\text{Layout}_1 = \left\{\mathcal{Z}_1^{1,\ldots,N}, \ldots, \mathcal{Z}_L^{P,\ldots,K}\right\} \tag{7}$$

while in the odd time steps, all the grids *slide* forward a column in order, forming the following new layout:

$$\text{Layout}_2 = \left\{\mathcal{Z}_0^{1,\ldots,N-n}, \mathcal{Z}_1^{N-n+1,\ldots,2N-n}, \ldots, \mathcal{Z}_L^{K-n,\ldots,K}\right\} \tag{8}$$

As shown in Figure 4, taking 2×2 grid for example, the grid sliding manipulation allows the two layouts to alternate repeatedly between the odd and even time steps of the diffusion process.

Through such a simple yet effective manipulation, the back-and-forth grid sliding indirectly achieves inter-grid interactions, encouraging the connections across multiple neighboring grids and thus maintaining long-term consistency throughout the video. Besides, existing video editing method [14, 16, 17, 20] store all data in GPU memory, which has a clear upper limit of processable video length. In contrast, the grid operations break up the whole video into parts, which makes it possible to employ a memory-swapping strategy that transfers

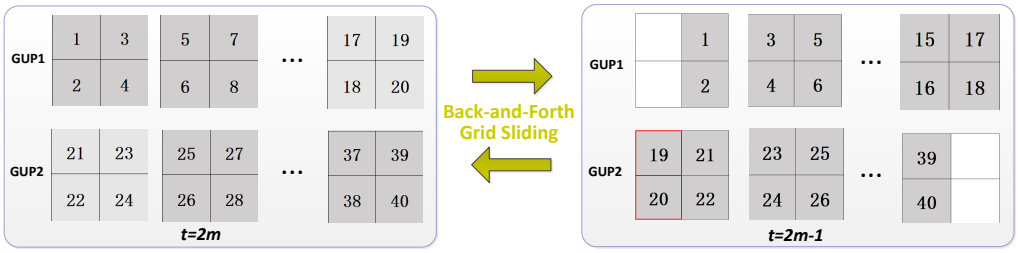


Figure 5: **The illustration of interaction between two GPUs**

temporarily unused data from GPU to disk space while transferring required data from disk back to GPU, theoretically enabling our UELVE to flexibly edit videos of arbitrary length using only a single GPU.

3.5. Multi-GPU Data Parallel Computing

In fact, when editing long videos using only a single GPU, compared to the grid diffusion with sliding manipulation, the time cost of memory swapping is more expensive, which still slows down the running speed. To enhance the efficiency, we further extend the grid operations to a multi-GPU parallel computing environment.

Specifically, given G GPUs, to process a video of K frames in parallel, the entire frame sequence may be uniformly divided into G clips, with each GPU assigned one containing a subsequence of $M = K/G$ frames. If existing methods [16, 17, 20] are extended to such multi-GPU case, each GPU will independently performs video editing task on its assigned clip, but the lack of cross-GPU interaction will result in noticeable visual discontinuities between the clips of the edited video. In contrast, our method, with fragmented grids, is easy to be deployed to multiple GPUs and the back-and-forth sliding manipulation between odd and even time steps in the diffusion process makes the interaction between

clips easy by just transferring only several latent frames at the boundary of every two adjacent GPU clips.

More precisely, as shown in Figure 5, when switching from an even time step (Layout_1) to an odd time step (Layout_2), the last n latent frames of GPU g are transferred to GPU $g+1$, where $g = 1, 2, \ldots, G-1$. When switching to the next even time step, these boundary frames would return to their original GPU. This ensures that latent frames near GPU boundaries continually participate in the attention computations of two adjacent GPUs, thereby maintaining visual and temporal continuity throughout the entire video. Leveraging a small scale of cross-GPU data communication at each diffusion timestep, the computational efficiency of our mehtod will increase by nearly G times, simultaneously ensuring the long-term consistency across GPUs.

4. Experiments

4.1. Experiment Settings

4.1.1. Datasets

The videos in our experiments source from the LongV-EVAL dataset [20], the VE-Bench dataset[43], AI-generated videos, and self-captured real-world videos. The LongV-EVAL dataset contains 75 video clips with an average duration of one minute, a resolution of 1280×720, and the frame rates ranging from 20 to 30 fps. The VE-Bench dataset contains 169 video sequences, each sequence has a fixed length of 32 frames, while the resolutions are not unified. For the AI-generated videos, we generate video samples via GLM-Qingying [44] using self-designed text prompts. Such videos have an average duration of one minute, a frame rate of 25 fps, and a resolution of 512×512. As for self-captured real-world videos, we record them with mobile phones in natural scenarios. They

range from 1 to 3 minutes in duration, with a frame rate of 30 fps and a resolution of 1920×1080. These videos cover diverse subjects, including humans, wildlife and natural landscapes. The text prompts corresponding to each video are collected from three sources: outputs from ChatGPT [45], captions adopted from prior studies, and manually designed textual descriptions. We resize the video frames to a uniform resolution of 512×512 before feeding them into models.

4.1.2. Implementation Details

We adopt Stable Diffusion v1-5 from the official Hugging Face Hub[46] as the pre-trained T2I model, and utilize DeepSeek [47] as the semantic analysis LLM. For the inversion and denoising diffusion process, we adopt the fixed classifier-free guidance scale of 12.5 with 50 timesteps. When fine-tuning is needed, we use the Adam optimizer [25] with a learning rate of 3×10^{-5} and train the model for 300 timesteps. Several baseline methods including RAVE [14], Tokenflow [13], Adaflow[20], and GLFM [48] support negative prompts, while other baseline methods lack this capability. To guarantee a fair comparison, negative prompts are disabled for all methods in our experiments. Our method incorporates two different control strategies from the existing image editing approaches, the attention control [26] scheme and the ControlNet-based [27] control scheme, to control the frame generation in video editing. In order to show that our method supports long-form video editing with low memory consumption and computation cost, all experiments are performed on NVIDIA RTX 4090 GPUs.

4.1.3. Baselines

We compare our UELVE method with several state-of-the-art baselines. To present the diversity of comparison for different editing types and goals, these

baselines include both zero-shot methods (such as FateZero [15], Control-A-Video [49], RAVE, TokenFlow, AdaFlow, and GLFM) and one-shot fine-tuning-based methods (such as Tune-A-Video [16], Video-P2P [17], and ControlVideo [10]). We can also categorize these methods according to their image editing techniques: ControlNet-based [27] approaches, including RAVE, ControlVideo, Control-A-Video and GLFM, and the attention control [26] based methods, including Tune-A-Video, Video-P2P, FateZero, TokenFlow and AdaFlow. Some of them perform favorably for global editing tasks, while others achieve better results on local editing. For all baseline methods, we adopt the default configurations from their official implementations to guarantee consistency and fair comparisons.

4.1.4. Metrics

We evaluate UELVE and the baseline methods using various video editing related metrics. The Structural Similarity Index Measure (SSIM) [50] quantifies the structural similarity between frames of the source and edited videos, focus on luminance, contrast, and structure. The Clip-T [14] evaluates semantic alignment between edited video and the target prompt by calculating the average distance of the CLIP [51] embedding of them. Tem-Con [15] assesses the continuity of features between consecutive video frames by computing the average cosine similarity of CLIP visual embeddings between adjacent frames. Masked Peak Signal-to-Noise Ratio(Masked PSNR) [17] evaluates the background preservation capacity by computing the Peak Signal-to-Noise Ratio (PSNR) [52] in background regions between the source video and its edited counterpart.

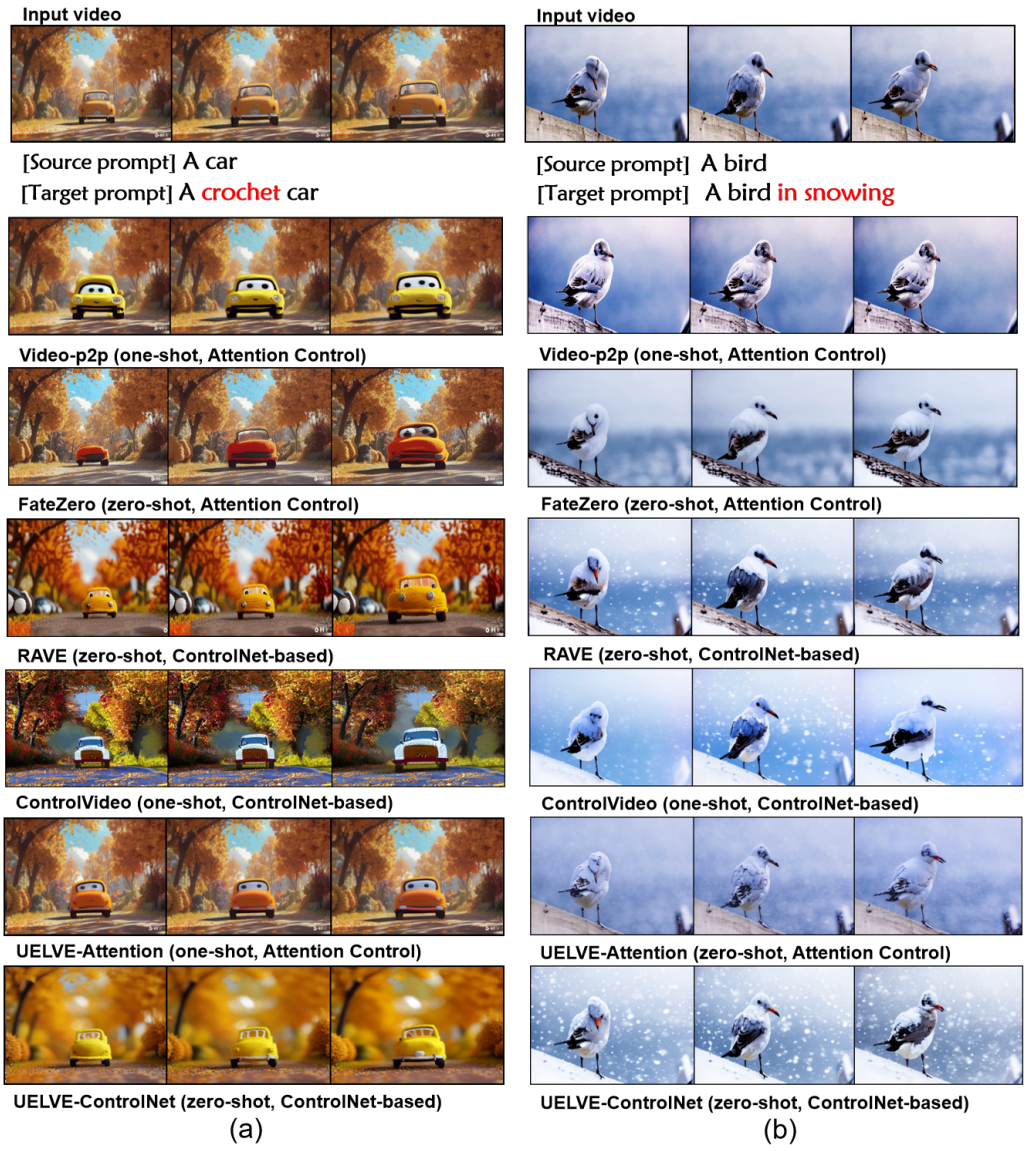


Figure 6: Qualitative comparison with baseline methods.

4.2. Comparisons With the State-of-the-art Methods

4.2.1. Qualitative Comparisons

Figure 6 shows some examples of editing results of baseline methods. The ControlNet-based methods RAVE and ControlVideo struggle with the local edit-

ing task (a) as they lack a mask mechanism, which often leads to severe corruption of background elements such as the trees, road, and bushes. By contrast, attention control methods Video-P2P and FateZero achieve favorable performance on (a) benefited from mask mechanisms, yet they exhibit inferior performance on global editing task (b), e.g., inadequate rendering of snowy style. Zero-shot methods FateZero and RAVE perform well on the simple global editing task (b) but struggle with the complex local editing task (a). Such as, FateZero fails to maintain consistency for the car and RAVE loses leaf and bush textures. In comparison, one-shot methods Video-P2P and ControlVideo strengthen editing ability by fine-tuning. Despite they perform well on both (a) and (b), fine-tuning incurs substantial time overhead. Constrained by static model configurations, most baseline methods excel at either complex local editing task (a) or simple global editing task (b). However, our UELVE method can adapt the model according to text prompts. It achieves favorable performance on both (a) and (b), while eliminating the unnecessary time overhead induced by redundant fine-tuning operation on simple editing task (b). Moreover, the performance of UELVE significantly outperform baseline methods when adopting the identical image editing strategies. These results not only demonstrate the capability of our unified video editing framework to flexibly handle diverse video editing tasks, but also verify the plug-and-play property of our UELVE method.

4.2.2. Quantitative Comparisons

Effectiveness Comparisons

Table 1 shows the quantitative evaluation results across all methods. Tune-A-Video, Video-P2P, FateZero, ControlVideo, and Control-A-Video process all frames at once. Limited by GPU memory capacity, such paradigm struggle

Table 1: Quantitative comparison with baseline methods. Bold indicates the best choice, and underline indicates the second best choice. ↑ denotes that higher values indicate better performance. "—" denotes that the method fails to process videos with such long durations.

Method	**Masked PSNR↑**				**SSIM↑**				**Clip-T↑**				**Tem-Con↑**			
	8-f	16-f	48-f	200-f	8-f	16-f	48-f	200-f	8-f	16-f	48-f	200-f	8-f	16-f	48-f	200-f
Tune-A-Video [16]	13.91	15.87	15.13	—	0.4350	0.4749	0.4418	—	0.2960	0.2727	0.2894	—	0.9488	0.9481	0.9441	—
Video-P2P [17]	23.66	22.23	—	—	0.6563	0.6217	—	—	0.2937	0.2955	—	—	0.9695	0.9640	—	—
FateZero [15]	23.83	—	—	—	0.6572	—	—	—	0.2872	—	—	—	0.9576	—	—	—
RAVE [14]	22.98	20.18	20.24	20.16	0.5680	0.5349	0.5894	0.5419	0.3145	0.3128	0.3044	0.2928	0.9795	0.9753	0.9696	0.9619
ControlVideo [10]	20.60	19.97	—	—	0.5923	0.5517	—	—	0.3070	0.3012	—	—	0.9746	0.9732	—	—
Control-A-Video [49]	10.72	9.715	11.52	—	0.4802	0.3944	0.4245	—	0.2942	0.3065	0.2979	—	0.9673	0.9638	0.9506	—
TokenFlow [13]	23.55	20.71	20.03	20.38	0.6502	0.6431	0.6313	**0.6130**	0.2917	0.2980	0.2973	0.2898	0.9776	0.9712	**0.9724**	0.9718
AdaFlow [20]	23.10	21.15	20.15	19.32	0.6500	0.6499	**0.6357**	0.5819	0.2945	0.2961	0.3040	0.2708	0.9789	0.9762	0.9720	0.9517
GLFM [48]	22.91	20.49	20.05	20.19	0.6031	0.6190	0.6217	0.6017	0.3149	0.3117	0.3037	**0.2991**	0.9791	0.9677	0.9626	0.9563
UELVE-Attention	**24.05**	**22.83**	**21.78**	**21.65**	**0.6849**	**0.6635**	0.6176	0.5833	0.2841	0.2955	0.2922	0.2893	0.9653	0.9655	0.9594	**0.9725**
UELVE-ControlNet	21.18	20.00	20.29	20.47	0.5711	0.5853	0.5412	0.5570	**0.3174**	**0.3210**	**0.3056**	0.2957	**0.9801**	**0.9765**	0.9707	0.9643

with long videos. Although RAVE, TokenFlow, AdaFlow and GLFM support long-video processing, they fail to simultaneously maintain short-term and long-term consistency, which compromises video editing performance. RAVE uses a shuffle strategy to maintain long-term consistency, but this scheme damages background content and results in low Masked PSNR and SSIM scores. TokenFlow and AdaFlow achieve competitive results on Tem-Con and SSIM scores, yet exhibit inferior performance on CLIP-T and Masked PSNR scores. This phenomenon stem from their strategy of maintaining consistency by propagating features from key frames to remaining frames. Moreover, AdaFlow adopts token pruning for long-video editing, causing editing quality decline. So, AdaFlow achieves inferior performance across all four metrics under the 200-f frames configuration. GLFM achieves favorable scores on SSIM and CLIP-T, while yielding weak results on Masked PSNR and Tem-Con. This indicates that it lacks competence in local editing tasks. With the same image control techniques, our UELVE surpasses baseline methods across most frame configurations. This con-

Table 2: The applicability for Multi-GPU data parallel in inversion and denoising stages, where ✓ indicates enable and × indicates disable. The inference time is the sum of inversion and denoising cost. For methods unable to process long videos due to GPU memory constraints, we denote their results with "—".

Method	Inversion	Denoising	8f	16f	48f	200f
Tune-A-Video [16]	×	×	19s	35s	1m42s	—
Video-P2P [17]	×	×	1m01s	2m04s	—	—
FateZero [15]	×	×	2m25s	—	—	—
RAVE [14] ×2GPU	✓	✓	19s	37s	1m44s	7m09s
ControlVideo [10]	×	×	1m02s	4m10s	—	—
Control-A-Video [49]	×	×	25s	46s	2m03s	8m12s
TokenFlow [13]	×	×	51s	1m48s	5m34s	21m41s
AdaFlow [20]	×	×	34s	1m12s	3m35s	15m01s
GLFM [48]	×	×	51s	1m55s	5m50s	21m06s
UELVE-Attention×2GPU	✓	✓	33s	1m03s	3m09s	13m45s
UELVE-ControlNet×2GPU	✓	✓	18s	34s	1m34s	6m12s

firms that our grid manipulation strategy effectively preserve both the short-term and long-term consistency.

Efficiency Comparisons

As shown in Table 2, we summarize the inference time overhead of all methods. Tune-A-Video, Video-P2P, FateZero, ControlVideo, Control-A-Video and GLFM adopt no acceleration strategies, leading to substantial time overhead. Among them, the ControlNet-based methods, e.g., ControlVideo, Control-A-Video and GLFM, run considerably faster than attention control methods such as Tune-A-Video, Video-P2P and FateZero. TokenFlow accelerates inference via token merging. Although it runs faster than the above unoptimized baselines, its acceleration gain remains limited and the overall runtime is still high. AdaFlow incorporates both token merging and token pruning for inference ac-

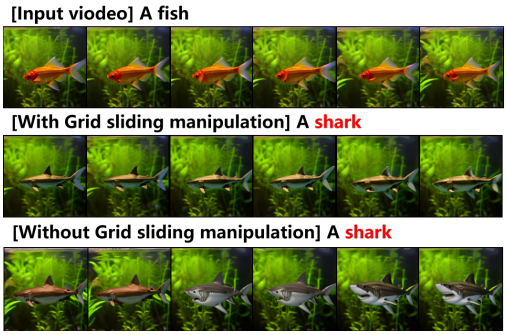


Figure 7: The ablation experiment of grid sliding manipulation. The shark's form maintain consistent, after our grid sliding manipulation.

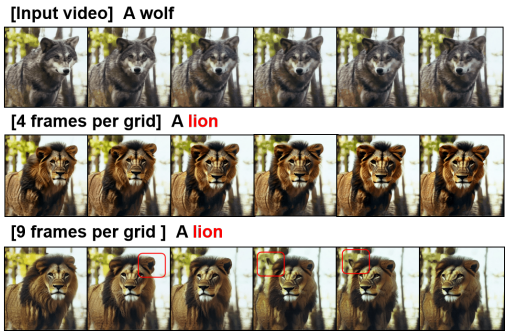


Figure 8: The ablation experiment of grid size. With the increase of grid size, the overall beauty is improved but the details are coarse

celeration. It consumes noticeably less time than TokenFlow. Since there are no publicly available multi-GPU data parallel video editing methods for comparison, we equip RAVE with our multi-GPU data parallel computing method to construct an accelerated baseline variant. Its inference speed is still inferior to our method. Our UELVE mostly achieves the fastest inference speed and significantly outperforms baseline methods when utilize the identical image editing control strategies. This observation demonstrates that our multi-GPU data parallel computing scheme can drastically accelerate video inference.

4.3. Ablation Studies

4.3.1. Grid Sliding Manipulation

We conduct an ablation experiment for the grid sliding manipulation. Sampling three grids from the grid sequences and select two frames within each grid to visualize the editing effects. As shown in Figure 7, without the grid sliding manipulation, the shark in different grids presents different shapes, while with the grid sliding manipulation, the shark between different grids present consistent shapes. It shows the necessity of our grid sliding manipulation in enhancing

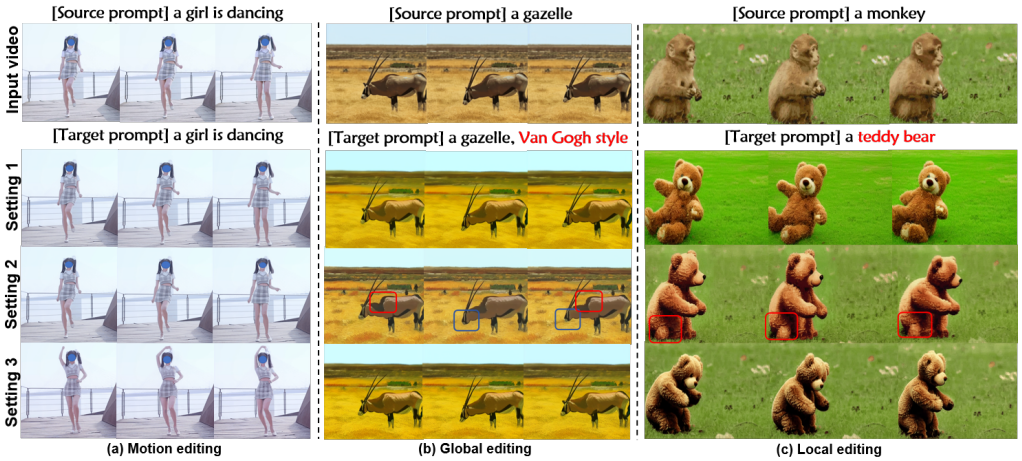


Figure 9: The ablation studies of LLM module in our unified video editing framework. The LLM module in our UELVE method adopts different configurations for diverse editing tasks: no mask with motion module fine-tuning for case (a), no mask without fine-tuning for case (b), and mask usage without fine-tuning for case (c).

inter-grid consistency and preserve the long term consistency.

4.3.2. The Selection of Grid Size

We also perform ablation studies on different grid sizes, as shown in Figure 8, a larger grid size enhances the model to better capture global context yet leads to coarser details. For instance, although a grid size of 9 (3×3) yields improved overall visual aesthetics and lion shape fidelity compared with a grid size of 4 (2×2), it also introduces prominent detail artifacts, such as a blurred dark left ear and an abnormally folded right ear in certain frames.

4.3.3. LLM's Effect

We further perform ablation experiments on the LLM module. Our UELVE utilizes an LLM for adaptive model adjustment according to text prompts, denoted as Setting 3. In the absence of the LLM module, only fixed model config-

urations (consistent with existing methods) are available. Specifically, setting 1 uses no masking mechanism without fine-tuning, setting 2 adopts a fixed masking mechanism and fine-tunes temporal attention. We evaluate all three settings on three different video editing tasks: motion editing in (a), global editing in (b), and local editing in (c). As shown in Figure 9, Setting 1 achieves favorable results for case (b). Nevertheless, it fails in case (c), where both the background and the motion of the monkey are drastically altered. Besides, it cannot complete case (a). Setting 2 loses partial details in case (b), such as the mouth and neck of the gazelle, and exhibits artifacts on the legs of the teddy bear for case (c). Similarly, it cannot handle case (a). By utilizing the LLM for adaptive model adjustment, Setting 3 successfully accomplishes case (a) and achieves competitive performance on both case (b) and case (c). These results indicate that fixed model configurations cannot accommodate diverse video editing tasks, which highlights the necessity of our LLM module in the proposed unified video editing framework.

4.3.4. Multi-GPU Acceleration

We also conduct ablation experiments on multi-GPU data parallel computing method. Since no open-source multi-GPU data-parallel video editing methods are available for comparison, we equip RAVE with our multi-GPU data parallel method to construct a new baseline. We also introduce Video-Infinity [23] as a baseline, which leverages multi-GPU data parallelism for long video generation. To preserve cross-GPU consistency, it broadcasts attention features for every GPU which incur substantial inter-GPU communication overhead. As shown in Table 3 , our UELVE achieves a significant improvement in editing speed with multi-GPU and delivers the best acceleration performance. Both RAVE and

Table 3: Ablation on multi-GPU data parallelism. Video-Infinity is a video generation method without an inversion stage. Since it processes all frames simultaneously, it cannot handle 200 frame videos on a single GPU due to GPU memory constraints. We use "—" to denote them.

Method	**8f**		**16f**		**48f**		**200f**	
	Inversion	Denoising	Inversion	Denoising	Inversion	Denoising	Inversion	Denoising
Video-Infinity[23]	—	12s	—	18s	—	48s	—	—
Video-Infinity[23] ×2GPU	—	22s	—	36s	—	40s	—	1m43s
RAVE[14]	17s	20s	30s	39s	1m23s	1m47s	5m22s	7m28s
RAVE[14] ×2GPU	9s	10s	17s	20s	47s	57s	3m08s	4m01s
UELVE-Attention	24s	34s	57s	1m01s	2m44s	3m16s	11m28s	14m25s
UELVE-Attention ×2GPU	15s	18s	29s	34s	1m28s	1m41s	6m06s	7m39s
UELVE-ControlNet	16s	18s	29s	36s	1m12s	1m38s	5m16s	6m07s
UELVE-ControlNet ×2GPU	9s	9s	16s	18s	40s	54s	2m50s	3m22s

Video-Infinity suffer degraded acceleration performance due to heavy inter-GPU communication. Notably, Video-Infinity even yields negative speedup ratios under small frame configurations, due to the heavy inter-GPU communication overhead.

5. Conclusion

In this paper, we present UELVE, an LLM-driven unified framework with grid diffusion for efficient long video editing. UELVE leverages an LLM to perform semantic analysis and reasoning to unify various video editing tasks. Based on grid diffusion with simple and effective grid manipulations, UELVE is able to maintain both short-term and long-term consistencies of the edited long-form video. In addition, UELVE is readily to be deployed on multiple GPUs in the data-parallelism manner, significantly improving the editing efficiency with only minor cross-GPU data communication. UELVE can be used as a plugin to combine various existing image editing techniques to achieve desirable video editing. Although the temporal awareness module enables the handling of simple motion editing tasks, effectively completing complex motion cases remains a challenge.

In future, we plan to incorporate specialized motion editing techniques to overcome this limitation.

Authorship contribution statement

Deyin Liu: Conceptualization, Methodology, Writing draft and Revising. **Li He**: Co-writing draft, Curating data, Visualization and Validation. **Zhan Li**: Formal Analysis and Validation. **Chengwu Liang**: Supervision, Review and Editing.

Acknowledgments

Thank the support of China Scholarship Council (CSC) for providing research fellowship for the first author. This work was also supported in part by the National Natural Science Foundation of China under Grant 62372150.

References

[1] W. Sun, R.-C. Tu, J. Liao, D. Tao, Diffusion model-based video editing: A survey, arXiv preprint arXiv:2407.07111 (2024).

[2] R. Rombach, A. Blattmann, D. Lorenz, P. Esser, B. Ommer, High-resolution image synthesis with latent diffusion models, in: CVPR, 2022, pp. 10674–10685.

[3] J. Chen, J. Yu, C. Ge, L. Yao, E. Xie, Z. Wang, J. T. Kwok, P. Luo, H. Lu, Z. Li, PixArt-alpha: Fast training of diffusion transformer for photorealistic text-to-image synthesis, in: ICLR, 2024.

[4] Y. Huang, J. Huang, Y. Liu, M. Yan, J. Lv, J. Liu, W. Xiong, H. Zhang, L. Cao, S. Chen, Diffusion model-based image editing: A survey, IEEE Transactions on Pattern Analysis and Machine Intelligence (2025) 4409–4437.

[5] D. J. Zhang, J. Z. Wu, J.-W. Liu, R. Zhao, L. H. Ran, Y. Gu, D. Gao, M. Z. Shou, Show-1: Marrying pixel and latent diffusion models for text-to-video generation, International Journal of Computer Vision (2023) 1879–1893.

[6] H. Chen, Y. Zhang, X. Cun, M. Xia, X. Wang, C. Weng, Y. Shan, VideoCrafter2: Overcoming data limitations for high-quality video diffusion models, in: CVPR, 2024, pp. 7310–7320.

[7] Y. HaCohen, N. Chiprut, B. Brazowski, D. Shalem, D. Moshe, E. Richardson, E. Levin, G. Shiran, N. Zabari, O. Gordon, et al., LTX-Video: Realtime video latent diffusion, arXiv preprint arXiv:2501.00103 (2025).

[8] A. Wang, B. Ai, B. Wen, C. Mao, C.-W. Xie, D. Chen, F. Yu, H. Zhao, J. Yang, J. Zeng, et al., Wan: Open and advanced large-scale video generative models, arXiv preprint arXiv:2503.20314 (2025).

[9] Z. Xing, Q. Feng, H. Chen, Q. Dai, H. Hu, H. Xu, Z. Wu, Y.-G. Jiang, A survey on video diffusion models, ACM Computing Surveys (2025) 41:1–41:42.

[10] M. Zhao, R. Wang, F. Bao, C. Li, J. Zhu, ControlVideo: Conditional control for one-shot text-driven video editing and beyond, Science China Information Sciences (2025) 132107:1–132107:13.

[11] Y. Lu, Y. Liang, L. Zhu, Y. Yang, FreeLong: Training-free long video generation with SpectralBlend temporal attention, NeurIPS (2024).

[12] Z. Jiang, Z. Han, C. Mao, J. Zhang, Y. Pan, Y. Liu, VACE: All-in-one video creation and editing, arXiv preprint arXiv:2503.07598 (2025).

[13] M. Geyer, O. Bar-Tal, S. Bagon, T. Dekel, TokenFlow: Consistent diffusion features for consistent video editing, in: ICLR, 2024, pp. 20637–20650.

[14] O. Kara, B. Kurtkaya, H. Yesiltepe, J. M. Rehg, P. Yanardag, RAVE: Randomized noise shuffling for fast and consistent video editing with diffusion models, in: CVPR, 2024, pp. 6507–6516.

[15] C. Qi, X. Cun, Y. Zhang, C. Lei, X. Wang, Y. Shan, Q. Chen, FateZero: Fusing attentions for zero-shot text-based video editing, in: ICCV, 2023, pp. 15932–15942.

[16] J. Z. Wu, Y. Ge, X. Wang, S. W. Lei, Y. Gu, Y. Shi, W. Hsu, Y. Shan, X. Qie, M. Z. Shou, Tune-A-Video: One-shot tuning of image diffusion models for text-to-video generation, in: ICCV, 2023, pp. 7589–7599.

[17] S. Liu, Y. Zhang, W. Li, Z. Lin, J. Jia, Video-P2P: Video editing with cross-attention control, in: CVPR, 2024, pp. 8599–8606.

[18] X. Li, C. Ma, X. Yang, M.-H. Yang, VidToMe: Video token merging for zero-shot video editing, in: CVPR, 2024, pp. 7486–7495.

[19] S. Tu, Q. Dai, Z.-Q. Cheng, H. Hu, X. Han, Z. Wu, Y.-G. Jiang, MotionEditor: Editing video motion via content-aware diffusion, in: CVPR, 2024, pp. 7882–7891.

[20] S. Zhang, Y. Liu, H. Zhou, J. Peng, Y. Zhou, X. Sun, R. Ji, AdaFlow: Efficient long video editing via adaptive attention slimming and keyframe selection, arXiv preprint arXiv:2502.05433 (2025).

[21] Z. Liu, Y. Wang, T. Wei, C. Ma, AnchorSync: Global consistency optimization for long video editing, arXiv preprint arXiv:2508.14609 (2025).

[22] F. Liu, S. Zhang, X. Wang, Y. Wei, H. Qiu, Y. Zhao, Y. Zhang, Q. Ye, F. Wan, Timestep embedding tells: It's time to cache for video diffusion model, in: CVPR, 2025, pp. 7353–7363.

[23] Z. Tan, X. Yang, S. Liu, X. Wang, Video-Infinity: Distributed long video generation, CoRR (2024). arXiv:2406.16260.

[24] Z. Chen, X. Ma, G. Fang, Z. Tan, X. Wang, AsyncDiff: Parallelizing diffusion models by asynchronous denoising, NeurIPS (2024) 95170–95197.

[25] D. P. Kingma, J. Ba, Adam: A method for stochastic optimization, arXiv preprint arXiv:1412.6980 (2014).

[26] A. Hertz, R. Mokady, J. Tenenbaum, K. Aberman, Y. Pritch, D. Cohen-Or, Prompt-to-prompt image editing with cross-attention control, in: ICLR, 2023, pp. 14369–14388.

[27] L. Zhang, A. Rao, M. Agrawala, Adding conditional control to text-to-image diffusion models, in: ICCV, 2023, pp. 3813–3824.

[28] W. Kong, Q. Tian, Z. Zhang, R. Min, Z. Dai, J. Zhou, J. Xiong, X. Li, B. Wu, J. Zhang, et al., Hunyuanvideo: A systematic framework for large video generative models, arXiv preprint arXiv:2412.03603 (2024).

[29] S. S. Harsha, A. Revanur, D. Agarwal, S. Agrawal, GenVideo: One-shot target-image and shape aware video editing using T2I diffusion models, in: CVPR Workshops, 2024, pp. 7559–7568.

[30] D. Liu, L. Y. Wu, X. Xie, Blended latent diffusion under attention control for real-world video editing, in: International Conference on Pattern Recognition and Artificial Intelligence, 2024, pp. 234–241.

[31] T. Brooks, A. Holynski, A. A. Efros, InstructPix2Pix: Learning to follow image editing instructions, in: CVPR, 2023, pp. 18392–18402.

[32] Y. Huang, L. Xie, X. Wang, Z. Yuan, X. Cun, Y. Ge, J. Zhou, C. Dong, R. Huang, R. Zhang, et al., SmartEdit: Exploring complex instruction-based image editing with multimodal large language models, in: CVPR, 2024, pp. 8362–8371.

[33] J. Mai, C. Wang, G. G. Qian, W. Menapace, S. Tulyakov, B. Ghanem, P. Wonka, A. Mirzaei, EasyV2V: A high-quality instruction-based video editing framework, arXiv preprint arXiv:2512.16920 (2025).

[34] Y. Zhang, Y. Wei, X. ZHANG, W. Zuo, Q. Tian, et al., Controlvideo: Training-free controllable text-to-video generation, in: International Conference on Learning Representations, Vol. 2024, 2024, pp. 54441–54461.

[35] J. Ho, A. Jain, P. Abbeel, Denoising diffusion probabilistic models, in: NeurIPS, 2020, pp. 6840–6851.

[36] J. Song, C. Meng, S. Ermon, Denoising diffusion implicit models, in: ICLR, 2021, pp. 14205–14225.

[37] T. Lee, S. Kwon, T. Kim, Grid diffusion models for text-to-video generation, in: CVPR, 2024, pp. 8734–8743.

[38] M. Li, L. Lin, Y. Liu, Y. Zhu, Y. Li, Qffusion: Controllable portrait video editing via quadrant-grid attention learning, IEEE Transactions on Visualization and Computer Graphics (2025).

[39] T. Yin, M. Gharbi, R. Zhang, E. Shechtman, F. Durand, W. T. Freeman, T. Park, One-step diffusion with distribution matching distillation, in: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition, 2024, pp. 6613–6623.

[40] J. Zhang, Y. Zhang, J. Gu, J. Dong, L. Kong, X. Yang, Xformer: Hybrid x-shaped transformer for image denoising, in: International Conference on Learning Representations, Vol. 2024, 2024, pp. 13333–13345.

[41] D. P. Kingma, M. Welling, Auto-encoding variational bayes, in: ICLR, 2014.

[42] X. Zhao, W. Ding, Y. An, Y. Du, T. Yu, M. Li, M. Tang, J. Wang, Fast segment anything (2023). arXiv:2306.12156.

[43] S. Sun, X. Liang, S. Fan, W. Gao, W. Gao, VE-Bench: Subjective-aligned benchmark suite for text-driven video editing quality assessment, in: AAAI, 2025.

[44] Zhipu AI, Qingying ai video generation, accessed: 2026-06-02 (2026). URL `https://chatglm.cn/video`

[45] OpenAI, Chatgpt [large language model], accessed: 2026-06-08 (2023). URL `https://chat.openai.com`

[46] R. Rombach, P. Esser, Stable diffusion v1-5, https://huggingface.co/stable-diffusion-v1-5/stable-diffusion-v1-5 (2022).

[47] L. Luo, L. Ma, L. Zhu, et al., Deepseek-v4: Towards highly efficient million-token context intelligence (2026). arXiv:2606.19348.

[48] Z. Yu, P. Qiao, J. Chen, X. Zhang, Joint global-local frames modeling to enhance semantic alignment for zero-shot long video editing, Neurocomputing (2025). doi:10.1016/j.neucom.2025.130836.

[49] W. Chen, Y. Ji, J. Wu, H. Wu, P. Xie, J. Li, X. Xia, X. Xiao, L. Lin, Control-A-Video: Controllable text-to-video diffusion models with motion prior and reward feedback learning, arXiv:2305.13840 (2023).

[50] Z. Wang, A. C. Bovik, H. R. Sheikh, E. P. Simoncelli, Image quality assessment: from error visibility to structural similarity, IEEE Trans. Image Process. (2004) 600–612.

[51] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, G. Krueger, I. Sutskever, Learning transferable visual models from natural language supervision, in: ICML, 2021.

[52] H. Jiang, D. Sun, V. Jampani, M. Yang, E. G. Learned-Miller, J. Kautz, Super slomo: High quality estimation of multiple intermediate frames for video interpolation, in: CVPR, 2018, pp. 9000–9008.